Cpts 317. 8/31

Last time: RegExp ← * , + , ·

loop, Choice, one is before another.

Elevator:

floor 3.

floor 2.

floor 1.

all the floor sequence when it travels from floor 1 back to floor 1.

Examples. 12321 ✓

12123232 1 ✓

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1 3 2 ✓ ended with floor 1.

floor 2 is skipped

is NOT valid.

ANSWER:

3 —

2 —

1 —

$$\underbrace{12(32+12)^{*}1}+1$$

short

test it: 12321 ✓

1321 ✗

Examples & Counter-Examples for a RegExp.

1^*0^*: 110 is an Exmple,

Λ is an Exmple,

011 is a Counter-Example.

$(10)^*$: Λ is an example,

1001 is a counter Example

1010 is an Example.

$((1+0)^*0)^*$: 10 is an Exmple,

1100 is an Exmple,

01 is a counter-Exmple

Let $\Sigma = \{0,1\}$.

1. all strings containing at least two 1's.

$$(0+1)^* \quad \underset{1}{\cancel{0}} \quad (0+1)^* \quad \underset{1}{\cancel{0}} \quad (0+1)^*$$

ans: $(0+1)^* 1 (0+1)^* 1 (0+1)^*$.

2. all strings containing exactly two 1's.

answer: $0^*10^*10^*$.

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substring: $ab\underbrace{baba}_{\text{substring}}a$

3. all strings containing substrings 101 and 010:

$$(0+1)^* \; 101 \; (0+1)^* \; 010 \; (0+1)^*$$

$+$

$$(0+1)^* \; 010 \; (0+1)^* \; 101 \; (0+1)^*$$

$+$

$$(0+1)^* \; 1010 \; (0+1)^*$$

$$(0+1)^* \quad 0101 \quad (0+1)^*$$

Don't try to simplify it.

4. All strings that do not end with 111.

Ans: What kind of substrings that we can end with, then?

you only have the following choices:

end with 0, OR

end with 01, OR

end with 011

Ans: $(0+1)^*0 + (0+1)^*01 + (0+1)^*011 +$

$\Lambda + 1 + 11$